

Redrob India Runs Hackathon – Dataset Investigation & Ranking Strategy Report

1. Executive Summary

This document summarizes our investigation into the Redrob India Runs Hackathon candidate-ranking challenge.

Rather than immediately building a ranking model, we first focused on understanding:

- The problem statement
- Evaluation expectations
- Dataset structure
- Candidate archetypes
- Behavioral signals
- Suspicious profiles and honeypots
- Hidden generation patterns

A full exploratory data analysis (EDA) and forensic audit were conducted on the 100,000-candidate dataset.

The investigation revealed that the dataset appears to be generated from a finite set of candidate archetypes and templates, with behavioral signals and profile consistency likely playing an important role in relevance determination.

The findings strongly suggest that successful ranking will require combining:

- Technical relevance to the JD
- Career-history evidence
- Behavioral signals
- Profile consistency

rather than relying solely on keyword matching.

2. Challenge Objective

The challenge requires ranking candidates for a specific AI/ML-focused job description.

The expected output consists of:

- Ranked candidates
- Fit scores
- Plain-language reasoning

The reasoning must:

- Reference actual profile facts
- Avoid hallucinations
- Be consistent with ranking
- Demonstrate understanding of the profile

The challenge documentation also explicitly warns about:

- Keyword stuffers
- Behavioral twins
- Plain-language Tier-5 candidates
- Honey pots

The CEO additionally advised participants to:

Look closely at the dataset.

This became the guiding principle of our investigation.

3. Investigation Strategy

Instead of building a model immediately, we divided the work into two stages.

Stage 1 – Dataset Reconnaissance

Objectives:

- Understand the candidate population
- Understand signal distributions
- Understand candidate archetypes
- Identify suspicious profiles
- Identify hidden patterns

Tasks completed:

- Schema audit
- Title analysis
- Skill analysis
- Company analysis
- Behavioral signal analysis
- Behavioral segmentation
- JD alignment exploration
- Honey pot exploration
- Candidate sampling

Stage 2 – Forensic Audit

Objectives:

- Validate suspicious profiles
- Detect generator artifacts
- Analyze category differences
- Reverse engineer candidate archetypes

Tasks completed:

- Consistency scoring
- Timeline validation
- Education validation
- Career-path validation
- Skill-role validation
- Generator artifact analysis

4. Dataset Overview

Dataset size:

100,000 candidates

Average values:

- Career history entries ≈ 3
- Skills ≈ 9.6
- Certifications are sparse
- India dominates candidate geography

Most common categories include:

- Business Analyst
- HR Manager
- Mechanical Engineer
- Accountant
- Project Manager
- Customer Support

Technical roles are present but significantly less common.

5. Behavioral Signal Findings

Behavioral signals appear surprisingly important.

Key signals include:

- recruiter_response_rate
- saved_by_recruiters_30d
- search_appearance_30d
- interview_completion_rate
- offer_acceptance_rate
- notice_period_days
- profile_views_received_30d
- github_activity_score

Observations:

- Search appearances correlate positively with recruiter saves.
- Search appearances correlate positively with salary expectations.
- Most correlations are weak, suggesting signals capture different dimensions.

Behavioral segmentation identified:

- Dormant candidates
- Passive candidates
- Moderately active candidates
- Highly active candidates

The highly active segment is extremely small.

This suggests engagement may be an important differentiator.

6. JD Alignment Findings

The JD focuses heavily on:

- Retrieval systems
- Ranking systems
- Recommendation systems
- Search infrastructure
- Embeddings
- Evaluation frameworks

However, EDA revealed:

Retrieval-related profiles are common.

Ranking-related profiles are comparatively rare.

This distinction may be extremely important.

Many candidates mention:

- RAG
- Vector search
- Embeddings

but far fewer mention:

- Ranking
- Learning-to-rank
- Relevance optimization

This suggests ranking-system experience may be a strong discriminator.

7. Candidate Archetypes Discovered

Several recurring archetypes emerged.

Archetype A – Genuine Retrieval / Ranking Engineers

Characteristics:

- Retrieval systems
- Learning-to-rank
- Recommendation systems
- Search infrastructure
- Product-company experience

Most likely high-value candidates.

Archetype B – Applied ML Engineers

Characteristics:

- NLP
- Embeddings
- LLMs
- Production ML

Strong candidates but not always retrieval-focused.

Archetype C – Data Engineers Transitioning to AI

Characteristics:

- Spark
- Airflow
- Warehousing
- Pipelines

Some AI skills but limited retrieval/ranking ownership.

Archetype D – AI Transition Candidates

Examples:

- Civil Engineer + LangChain
- Accountant + RAG
- Marketing Manager + LLMs

Typically have:

- Recent AI coursework
 - AI side projects
 - Weak production AI evidence
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Archetype E – Keyword Stuffers

Characteristics:

- Large AI skill inventory
- Minimal supporting work history

Likely intentional traps.

Archetype F – Consistency Failures

Characteristics:

- Timeline contradictions
- Title-description mismatches

- Career incoherence

Potential honeypots or generation artifacts.

8. Suspicious Candidate Investigation

The forensic audit identified several anomaly categories.

Most frequent:

- Previous title-description mismatch
- Skills-career mismatch
- Skills-role mismatch
- Current title-description mismatch
- Graduation after employment start
- Abrupt career jumps
- Salary inversions
- Signup-after-last-active violations

These issues were common enough that consistency became a major analytical signal.

9. Generator Artifact Findings

One of the most important discoveries.

The dataset appears heavily template-driven.

Examples include:

Repeated Summaries

Many summaries appear hundreds of times with only:

- years of experience
- title

changing.

Repeated Headlines

Examples:

- "HR manager helping teams scale"

- "Business analyst driving business outcomes"

appear thousands of times.

Repeated Role Descriptions

Some role descriptions occur more than 25,000 times.

Repeated Education Patterns

Identical degree/year combinations repeat frequently.

Repeated Career Paths

Many candidates share identical career trajectories.

Conclusion:

The dataset appears generated from archetype templates rather than 100,000 unique handcrafted profiles.

10. Hidden Signal Analysis

The forensic audit examined predictors of:

- AI-title status
- JD alignment
- Suspicion

Important predictors included:

JD Alignment

- Skill count
- Search appearances
- Recruiter saves
- Salary expectations
- Career-role count

Suspicion

- Skill count
- Consistency score
- Consistency reason count

This suggests that profile complexity and profile coherence both matter.

11. Lessons Learned

Several assumptions proved incorrect.

Incorrect Assumption

More AI keywords = better candidate.

Reality:

Many AI-heavy profiles lack supporting career evidence.

Incorrect Assumption

All JD matches are equally relevant.

Reality:

Retrieval-related skills are common.

Ranking-related skills are rare.

Incorrect Assumption

Behavioral signals are redundant.

Reality:

Behavioral metrics appear to capture independent dimensions.

12. Recommended Ranking Framework

The current evidence suggests a ranking system should combine:

Technical Fit (40%)

Evidence of:

- Retrieval
 - Ranking
 - Recommendation systems
 - Search infrastructure
 - Embeddings
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Career Evidence (25%)

Evidence from:

- Current role
- Previous roles
- Job descriptions

Not just skills.

Behavioral Signals (20%)

Signals such as:

- Recruiter response rate
 - Recruiter saves
 - Search appearances
 - Interview completion rate
 - Notice period
 - Open-to-work status
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Consistency Score (15%)

Penalties for:

- Timeline issues
 - Role-description mismatches
 - Career incoherence
 - Education inconsistencies
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13. Current Status

Completed:

- ✓ Dataset reconnaissance
- ✓ Behavioral analysis
- ✓ Candidate sampling
- ✓ JD alignment analysis
- ✓ Forensic audit
- ✓ Generator artifact analysis
- ✓ Suspicious profile investigation
- ✓ Archetype discovery

Pending:

- Final feature engineering
 - Ranking algorithm implementation
 - Reasoning generation framework
 - Submission generation
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14. Final Conclusion

The investigation indicates that the challenge is not primarily about semantic similarity.

It is about identifying the right candidate archetypes while avoiding misleading signals.

The strongest evidence currently suggests that successful ranking will depend on:

1. Retrieval/ranking-system relevance
2. Career-history evidence
3. Behavioral engagement signals
4. Candidate consistency

The dataset contains clear template structures, recurring archetypes, and deliberate distractions. Understanding these patterns before building the ranker has provided a strong foundation for the next stage of the project.